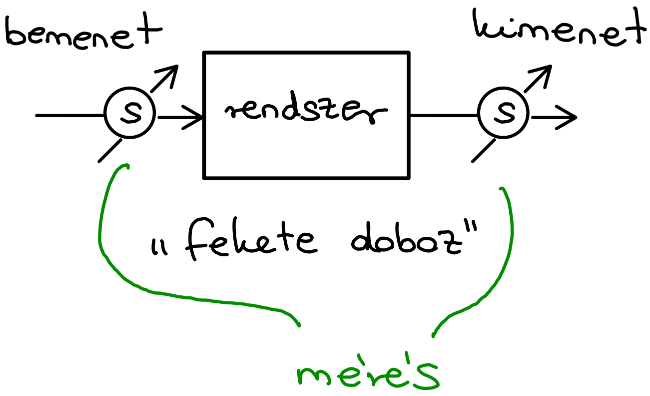
2. Előadás

RENDSZEREK DISZKRÉT ÉS FOLYTONOS IDEJŰ MODELLJEI

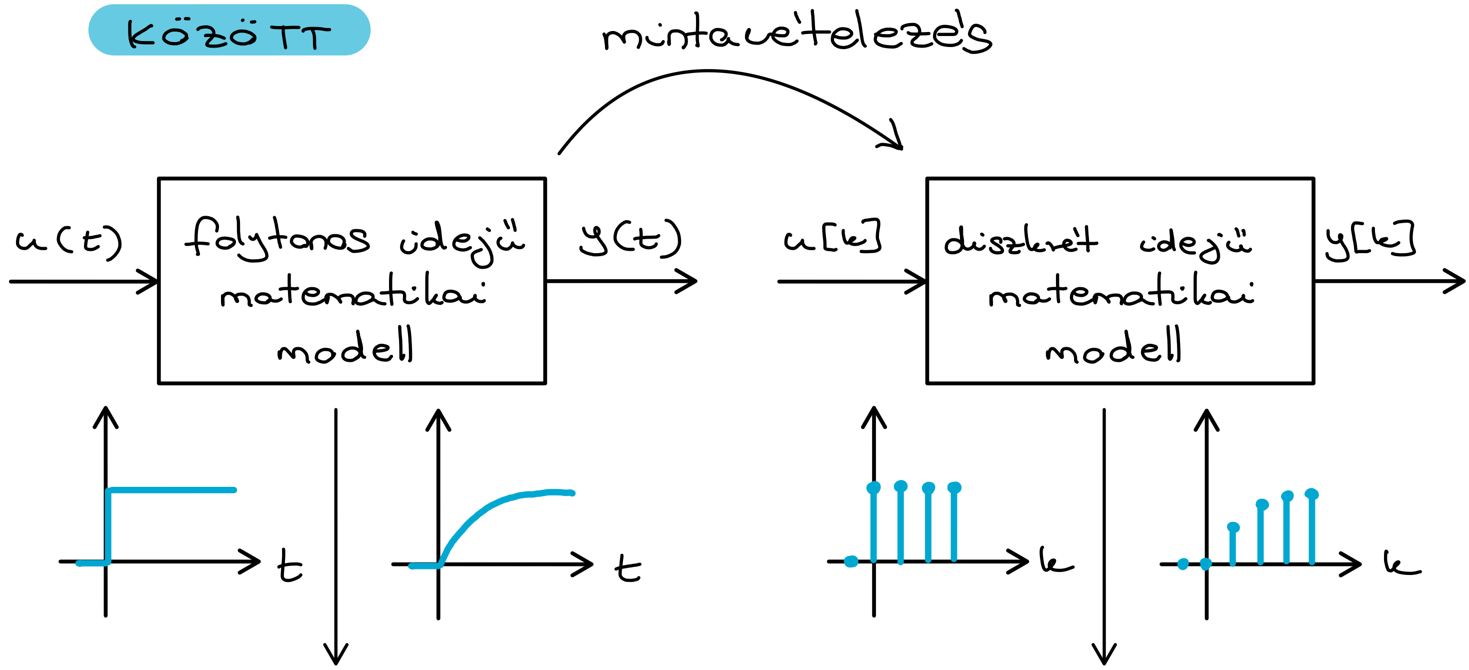


analóg: $t \in \mathbb{R}$

mintavételes: $k \in \mathbb{Z}$

adatgyűjtés szenzorokkal

2.1 KAPCSOLAT A FOLYTONOS ÉS DISZKRÉT MODELLEK KÖZÖTT



differenciálegyenlet

↳ 1 db n-ed rendű

↳ n db elsőrendű (ÁTM)

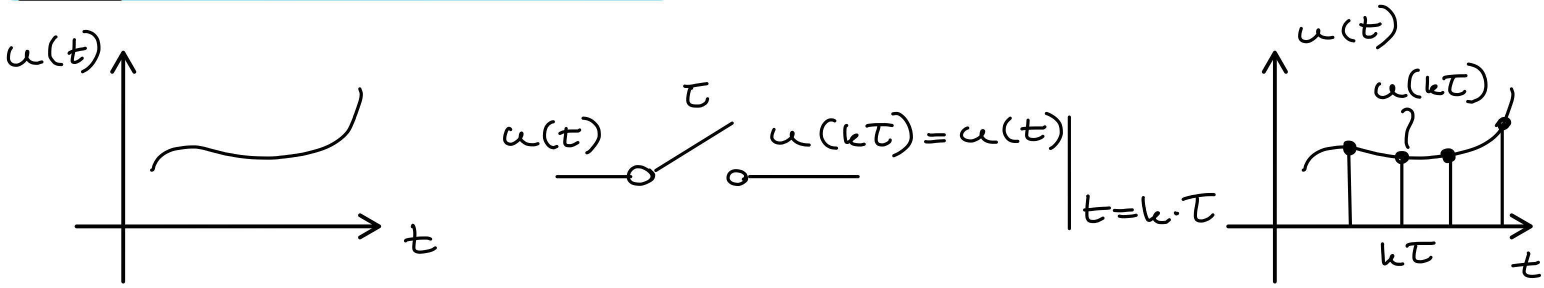
rekonstrukció

differenciaegyenlet

↳ 1 db n-ed rendű (ARMA)

↳ n db elsőrendű (ÁTM)

2.2 MINTAVÉTELEZÉS



mintavételezési függvény → számsorozat

τ – mintavételezési idő [s]

$(\Delta t, T_s, t_s, \ldots)$

$f = \frac{1}{\tau}$ – mintavételezési frekvencia [Hz]

$t_k = k \cdot \tau \longrightarrow u(k\tau) \longrightarrow$ tárolás

pl.: tömbben

	$u(0)$	$u(\tau)$	$u(2\tau)$	...
k:	0	1	2	...

↳ számsorozat k. eleme ($u[k]$)

2.3 DISZKRÉT IDEJŰ ÁLLAPOTTÉR MODELL

↳ rekurzív modell

$\underline{x}[k+1] = \underline{\underline{A}}_d\, \underline{x}[k] + \underline{\underline{B}}_d\, \underline{u}[k]$ – állapotegyenlet

$\underline{y}[k] = \underline{\underline{C}}_d\, \underline{x}[k] + \underline{\underline{D}}_d\, \underline{u}[k]$ – kimeneti egyenlet

↳ $k \in \mathbb{Z}$

$\underline{x} \in \mathbb{R}^n$: állapotvektor

$\boxed{\underline{u} \in \mathbb{R}^m}$: bemeneti vektor

$\boxed{\underline{y} \in \mathbb{R}^p}$: kimeneti vektor

↳ MIMO

SISO esetén $m \equiv p \equiv 1$

$\underline{\underline{A}} \in \mathbb{R}^{n \times n}$: rendszermátrix

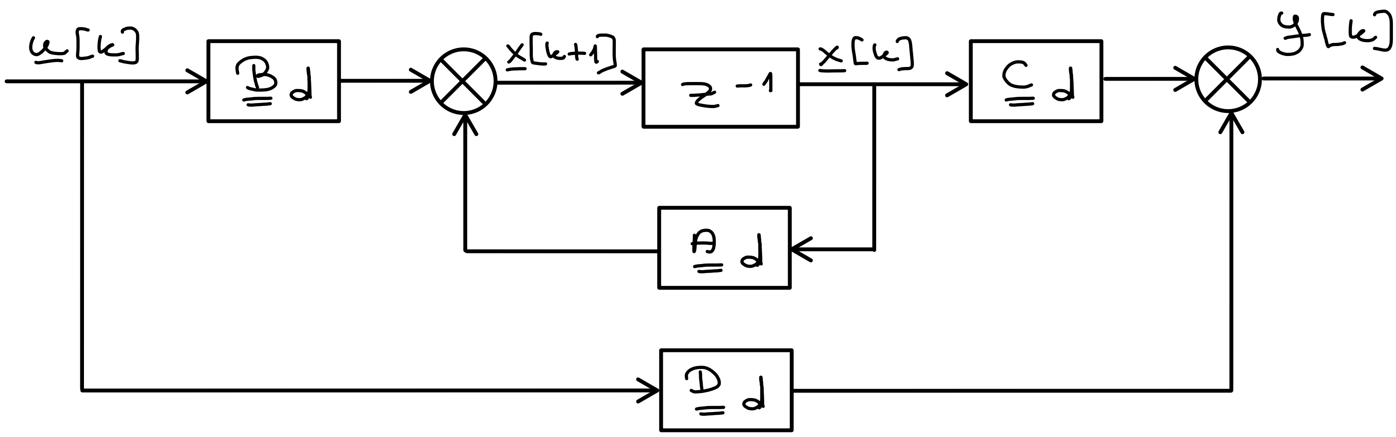
$\underline{\underline{B}} \in \mathbb{R}^{n \times m}$: bemeneti mátrix

$\underline{\underline{C}} \in \mathbb{R}^{p \times n}$: kimeneti mátrix

$\underline{\underline{D}} \in \mathbb{R}^{p \times m}$: előrecsatolási / segédmátrix

+ d : diszkrét idő jelölése

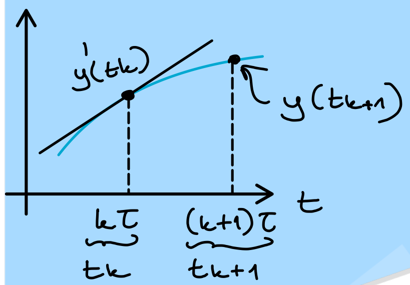
Hatásvázlat:



2.4 FOLYTONOS IDEJŰ ÁLLAPOTTÉR MODELL

$$\underline{x}[k+1] - \underline{x}[k] = \underline{\underline{A}}_d\, \underline{x}[k] + \underline{\underline{B}}_d\, \underline{u}[k] - \underline{x}[k] \quad \Big/ \cdot \frac{1}{\Delta t}$$

$$\frac{\underline{x}[k+1] - \underline{x}[k]}{\Delta t} = \frac{(\underline{\underline{A}}_d - \underline{\underline{I}})\,\underline{x}[k] + \underline{\underline{B}}_d\, \underline{u}[k]}{\Delta t}$$



$\underline{\underline{I}}$: n x n egységmátrix

$$\frac{\underline{x}(t_{k+1}) - \underline{x}(t_k)}{\Delta t} = \frac{1}{\Delta t} \cdot (\underline{\underline{A}}_d - \underline{\underline{I}})\,\underline{x}(t_k) + \frac{1}{\Delta t}\, \underline{\underline{B}}_d\, \underline{u}(t_k) \quad \Big/ \lim_{\Delta t \to 0}$$

$$y'(t_k) = \frac{y(t_{k+1}) - y(t_k)}{\underbrace{t_{k+1} - t_k}_{\sim \Delta t}}$$

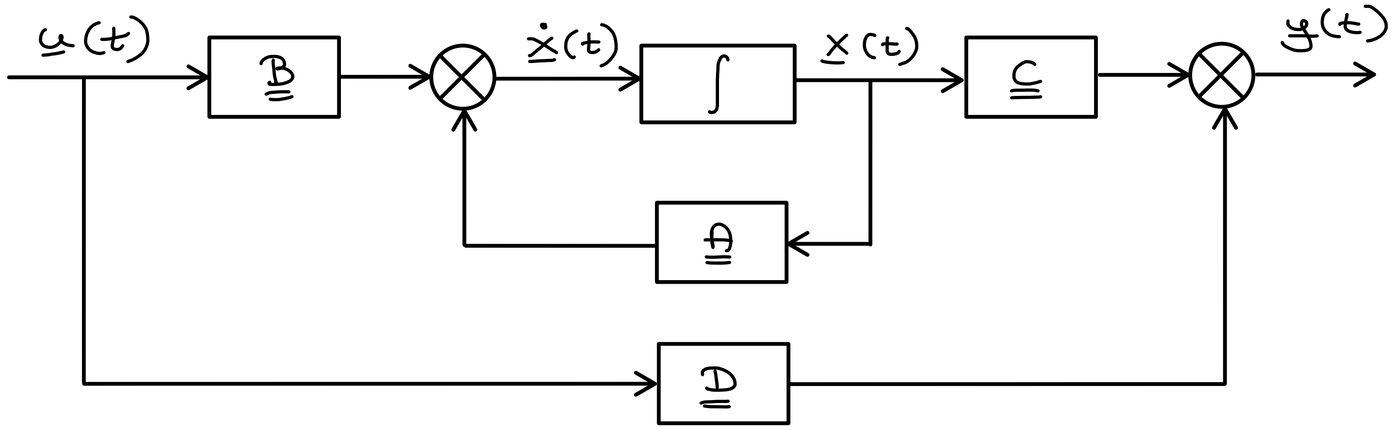
$$\lim_{\Delta t \to 0} \frac{1}{\Delta t} (\underline{\underline{A}}_d - \underline{\underline{I}}) = \underline{\underline{A}}$$

$$\lim_{\Delta t \to 0} \frac{1}{\Delta t} \underline{\underline{B}}_d = \underline{\underline{B}}$$

$\dot{\underline{x}}(t) = \underline{\underline{A}}\, \underline{x}(t) + \underline{\underline{B}}\, \underline{u}(t)$ – folytonos idejű állapotegyenlet

$\underline{y}(t) = \underline{\underline{C}}\, \underline{x}(t) + \underline{\underline{D}}\, \underline{u}(t)$ – folytonos idejű kimeneti egyenlet

Hatásvázlat:



2.5 DISZKRETIZÁLÁS

↳ A közelítéstől függ a diszkrét idejű modell alakja

- Előretartó Euler módszer (forward Euler method)

$$\underline{\underline{A}}_d = \underline{\underline{I}} + \underline{\underline{A}}\,\Delta t$$

$$\underline{\underline{B}}_d = \underline{\underline{B}} \cdot \Delta t$$

$$\underline{\underline{C}}_d = \underline{\underline{C}}$$

$$\underline{\underline{D}}_d = \underline{\underline{D}}$$

$$\dot{\underline{x}}(t) = \frac{\underline{x}(t_{k+1}) - \underline{x}(t_k)}{\Delta t}$$

$$\frac{\underline{x}(t_{k+1})}{\Delta t} - \frac{\underline{x}(t_k)}{\Delta t} = \underline{\underline{A}}\,\underline{x}(t_k) + \underline{\underline{B}}\,\underline{u}(t_k)$$

$$\underline{x}(t_{k+1}) = (\underline{\underline{A}} \cdot \Delta t + \underline{\underline{I}})\,\underline{x}(t_k) + \underline{\underline{B}}\,\Delta t\,\underline{u}(t_k)$$

$$\underline{y}(t_k) = \underline{\underline{C}}\,\underline{x}(t_k) + \underline{\underline{D}}\,\underline{u}(t_k)$$

- Hátratartó Euler módszer (backword Euler method)

$$\dot{\underline{x}}(t_k) \approx \frac{\underline{x}(t_k) - \underline{x}(t_{k-1})}{\Delta t} = \frac{\Delta \underline{x}(t_k)}{\Delta t}$$

$$\frac{\underline{x}(t_k) - \underline{x}(t_{k-1})}{\Delta t} = \underline{\underline{A}}\,\underline{x}(t_k) + \underline{\underline{B}}\,\underline{u}(t_k)$$

$$\underline{x}(t_k) = \underline{x}(t_{k-1}) + \Delta t\,\underline{\underline{A}}\,\underline{x}(t_k) + \Delta t\,\underline{\underline{B}}\,\underline{u}(t_k) \quad \Big/ - \Delta t\,\underline{\underline{A}}\,\underline{x}(t_k)$$

$$(\underline{\underline{I}} - \Delta t\,\underline{\underline{A}})\,\underline{x}(t_k) = \underline{x}(t_{k-1}) + \Delta t\,\underline{\underline{B}}\,\underline{u}(t_k)$$

$$\Longrightarrow \underline{x}(t_k) = (\underline{\underline{I}} - \underline{\underline{A}}\,\Delta t)^{-1}\,\underline{x}(t_{k-1}) + (\underline{\underline{I}} - \underline{\underline{A}}\,\Delta t)^{-1}\,\Delta t\,\underline{\underline{B}}\,\underline{u}(t_k)$$

A'TM alak:

$$\underline{x}(t_{k+1}) = \underline{\underline{A}}_d\, \underline{x}(t_k) + \underline{\underline{B}}_d\, \underline{x}(t_k)$$

„tömb" :

k:	0	1	2	...
	$x(t_{k-1})$	$x(t_k)$	$x(t_{k+1})$	...

↳ új állapot : $\underline{w}[k] = \underline{x}[k-1]$

$$\underline{w}[k+1] = \underbrace{(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1}}_{\underline{\underline{A}}_d} \underline{w}[k] + \underbrace{(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} \Delta t\, \underline{\underline{B}}}_{\underline{\underline{B}}_d}\, \underline{u}[k]$$

$$\underline{y}(t_k) = \underline{\underline{C}}_d \boxed{\underline{x}(t_k)} + \underline{\underline{D}}_d\, \underline{u}(t_k)$$

$$\underline{x}(t_k) = \underline{w}(t_{k+1}) = \underline{\underline{A}}_d\, \underline{w}[k] + \underline{\underline{B}}_d\, \underline{u}[k]$$

$$\underline{y}(t_k) = \underline{\underline{C}}(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} \underline{x}(t_{k-1}) + \underline{\underline{C}}(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} \Delta t\, \underline{\underline{B}}\, \underline{u}(t_k) + \underline{\underline{D}}\, \underline{u}(t_k)$$

$$\underline{y}[k] = \underline{\underline{C}}_d \cdot \underline{w}[k] + \underbrace{\left(\underline{\underline{C}}(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} \Delta t\, \underline{\underline{B}} + \underline{\underline{D}}\right)}_{\underline{\underline{D}}_d} \underline{u}[k]$$

⇒

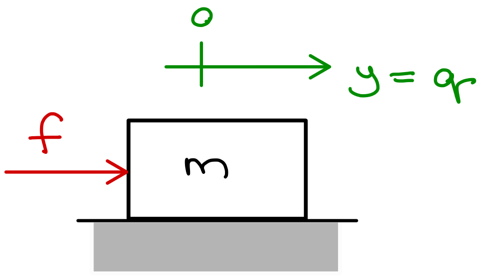
$$\underline{\underline{A}}_d = (\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1}$$

$$\underline{\underline{B}}_d = (\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} \Delta t\, \underline{\underline{B}}$$

$$\underline{\underline{C}}_d = \underline{\underline{C}}(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} = \underline{\underline{C}}\,\underline{\underline{A}}_d$$

$$\underline{\underline{D}}_d = \underline{\underline{C}}(\underline{\underline{I}} - \underline{\underline{A}}\Delta t)^{-1} \Delta t\, \underline{\underline{B}} + \underline{\underline{D}} = \underline{\underline{C}}\,\underline{\underline{B}}_d + \underline{\underline{D}}$$

2.6 PÉLDA: ANYAGI PONT MOZGÁSA



kimenet: $y = q$ (általánosított koordináta; itt: elmozdulás)

bemenet: $u = f$

$$m \cdot a(t) = f(t)$$

Állapotok: $x_1(t) = q(t)$

$$x_2(t) = v(t)$$

$$\underbrace{\begin{aligned} m a(t) &= f(t) \\ a(t) &= \dot{v}(t) \\ v(t) &= \dot{q}(t) \end{aligned}}$$

Folytonos idejű modell

$$\underbrace{\begin{bmatrix} \dot{q}(t) \\ \dot{v}(t) \end{bmatrix}}_{\underline{\dot{x}}(t)} = \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\underline{\underline{A}}} \underbrace{\begin{bmatrix} q(t) \\ v(t) \end{bmatrix}}_{\underline{x}(t)} + \underbrace{\begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix}}_{\underline{\underline{B}}} \underbrace{f(t)}_{u(t)}$$

$$y(t) = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\underline{\underline{C}}} \begin{bmatrix} q(t) \\ v(t) \end{bmatrix} + \underbrace{0}_{\underline{\underline{D}}} \cdot u(t)$$

$$\Downarrow \text{ diszkretizálás}$$

Diszkrét idejű modell:

1.) Előretartó Euler

$$\underline{\underline{A}}_d = \underline{\underline{I}} + \underline{\underline{A}} \Delta t = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \Delta t \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

$$\underline{\underline{B}}_d = \Delta t \, \underline{\underline{B}} = \begin{bmatrix} 0 \\ \Delta t / m \end{bmatrix}$$

$$\underline{\underline{C}}_d = \underline{\underline{C}} \qquad \underline{\underline{D}}_d = \underline{\underline{D}}$$

2.) Hátratartó Euler

$$(\underline{\underline{I}} - \underline{\underline{A}}\,\Delta t)^{-1} = \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & \Delta t \\ 0 & 0 \end{bmatrix} \right)^{-1} =$$

$$= \begin{bmatrix} 1 & -\Delta t \\ 0 & 1 \end{bmatrix}^{-1} = \underbrace{\frac{1}{1}}_{\frac{1}{\det(\cdot)}} \cdot \underbrace{\begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}}_{\mathrm{adj}(\cdot)}$$

$$\underline{\underline{A}}_d = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}$$

$$\underline{\underline{B}}_d = \underline{\underline{A}}_d\,\Delta t\,\underline{\underline{B}} = \begin{bmatrix} \Delta t^2 / m \\ \Delta t / m \end{bmatrix}$$

$$\underline{\underline{C}}_d = \underline{\underline{C}}\,\underline{\underline{A}}_d = \begin{bmatrix} 1 & \Delta t \end{bmatrix}$$

$$\underline{\underline{D}}_d = \underline{\underline{C}}\,\underbrace{\underline{\underline{A}}_d\,\Delta t\,\underline{\underline{B}}}_{\underline{\underline{B}}_d} + \underline{\underline{D}} = \Delta t^2 / m$$